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A MAGNETOTECTONIC STUDY OF THE HERCYNIAN MONTAGNE NOIRE (FRANCE)

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Abstract. A paleomagnetic study of eleven sites of Devonian limestones sampled along the flanks and hinge of a polyphase Hercynian recumbent fold in the Montagne Noire reveals well defined directions of characteristic remanent magnetization (ChRM). The directions have good within-site precision but a large between-site scatter, which fails to be reduced by a classical tilt correction. This is readily understood if widespread remagnetization occurred between the two major Hercynian phases of folding (A and B), which are known to have shaped the Montagne Noire. This remagnetization appears to be an instance of a continent-wide chemical event that wiped out much of the magnetic memory of Devonian and lower Carboniferous rocks in Laurasia, Baltica and northern Gondwana, possibly due to groundwater circulation following the major Hercynian tectonic phases. Comparison of observed and predicted Hercynian magnetic directions allows one to determine, through a fairly simple geometric construction, the tectonic elements of the later phase B. It is found that different rotations affected the flanks and hinge zone of the folds generated by phase A, thus demonstrating how a pre-existing structure can control the development of deformation in a later tectonic phase. The effects of the non cylindrical phase B can next be removed to reveal the broad tectonic features of phase A deformations. The reversed (Kiaman) polarity of the ChRM allows a determination of the age of phase B, which is found to be about 300-310 Ma and of the age of the remagnetization itself at about 310-320 Ma. The search for the age of remagnetization included paleomagnetic measurements at two lower Permian sites from the nearby Lodève basin. These provide new data for the Permian pole of Europe and point out the limitations of some earlier studies, due to insufficient magnetic cleaning. Finally, the principal shortening directions of the two phases can be determined from the magnetic data and are found to have rotated counterclockwise by 30° between phase A and phase B. This feature, already observed in the collision of India against Eurasia, may be related to the progressive suturing of former subduction zones during the Hercynian collision of Gondwana against Laurasia. This study demonstrates the potential of remagnetizations for the solution of rather complex tectonic problems. We propose the term "magnetotectonics" for this

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rapidly expanding field of studies, in which the methods of paleomagnetism and structural analysis are closely associated.

1. INTRODUCTION

Paleomagnetism has long been recognized as a potentially powerful tool to study the tectonic evolution of orogenic belts (for a recent review see Van der Voo and Channell [1980]). Until recently, however, tectonic analysis of paleomagnetic data almost exclusively relied on Graham's [1949] classical unfolding by rotation of the observed bedding plane back to horizontal about its line of strike. Yet, it is well known that both folding with plunging axes and interference of multiple phases of folding lead to rotations about inclined axes [e.g., Ramsay, 1967]. The corresponding discrepancy between net and apparent tectonic rotations has been emphasized by McDonald [1980]. Restoring for instance the curved termination of a plunging syncline to horizontal with Graham's scheme violates what is known about the acquisition of such a complicated shape. Recent progress has required a closer integration of techniques from paleomagnetism and structural analysis. For instance, Bonhommet et al [1981] convincingly unravelled two interfering phases of folding in the Asturian arc (Spain) and recovered the direction of a badly scattered primary remanence that could not be determined with the classical tilt correction. Using the magnetic remanence as a deformation marker, Perroud and Cobbold [1984] suggested, again in Asturia, that a bending about a vertical axis was probably responsible for the development of the hinge zone of a plunging fold. In the two above studies, the primary remanence had not been erased by later tectono-thermal events. Yet, the discovery of partial, or worse total, remagnetization, which often comes as a disappointment, is not infrequent in paleomagnetic studies. Still, McClelland-Brown [1983] was able to use two discrete episodes of remagnetization in folds of Old Red Sandstone, in which the plunge varied with time, to reconstruct intermediate stages of their evolution. Kligfield et al. [1983] and Cogné and Perroud [1985] studied Permian beds from the Alpes Maritimes in which not only rigid rotations but also plastic strain had to be taken into account. We present here the results from a magnetotectonic study of Paleozoic limestones in which total remagnetization occurred in the midst of Hercynian tectonic phases. Despite this, detailed combined magnetic and tectonic analyses lead to significant improvement in understanding of the tectonic history of the area.

2. GEOLOGICAL SETTING

The Montagne Noire, located at the southern tip of the Massif Central (France), is one of the southernmost sections of the western European Hercynian belt (Figure 1). It is only mildly affected by Pyrenean tectonics. The massif displays a rough ENE-WSW cylindrical symmetry (Figures 1a, 1b) with an axial zone of gneisses and Hercynian granites. It is bordered to the north by overlapping SE vergent thrust slices of Cambrian to Silurian sediments, and to the south by a pile of nappes, consisting of Cambrian to Visean sediments which form large south verging recumbent folds [Arthaud, 1970]. The paleomagnetic sites have been sampled in very well dated sediments of lower (Emsian) to upper (Famennian) Devonian age [Feist, 1983; R. Feist, personal communication, 1984). Most sections were fairly densely sampled, originally for magneto-stratigraphic purposes, mostly in characteristic red to gray griotte limestones of lower Famennian age (goniatite fossils).

The competent Devonian strata are part of a series going from (at least) Ordovician to lower Carboniferous (Visean). The area of sampling, termed the Roquebrun "false syncline" or synform (Figures 1a, 1c) results from a complicated series of Hercynian tectonic phases which have been carefully described both from a large scale and from a microtectonic point of view by Arthaud [1970]. The synform is part of the Montpeyroux nappe, one of a series of nappes that were thrusted from the North over the axial zone of the Montagne Noire. Arthaud recognizes four phases of folding. Phase 1 is the major one and has led to the formation of large E-W trending recumbent folds with flat-lying (S1) cleavage. Phase 2 is less intense and increases the amplitude of phase 1 folds. Phase 3 corresponds to southward overturning of the structures, small-scale folding and a new weakly south-dipping cleavage (S3). These three phases are roughly homo-axial and can be combined into a single one which we will term here phase A in order to avoid confusion. This is followed by a

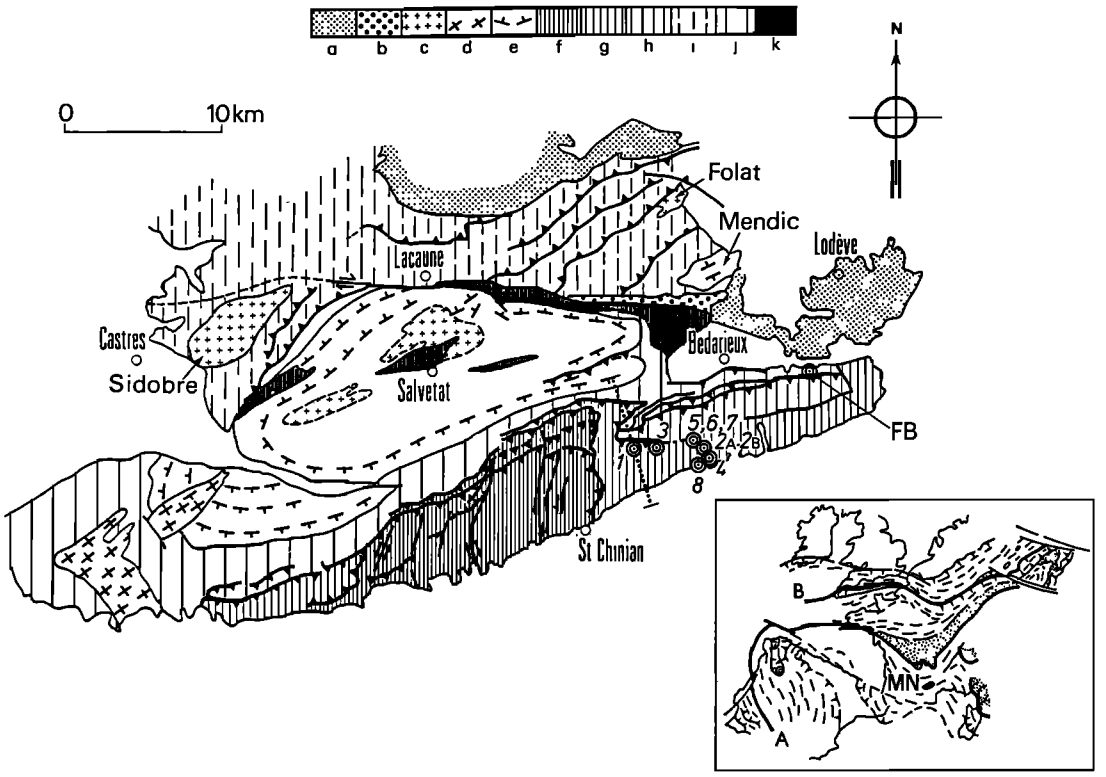


Fig. 1a. Geological setting of the Montagne Noire. Simplified geological map from Arthaud [1970]. Inset shows approximate locations of proposed sutures in the Hercynian orogenic belt [after Matte, 1986] and the study area (MN). Key to the geologic symbols: a, Permian, b, Stephanian (U. Carboniferous); c, Post-tectonic granites; d, Deformed Hercynian granites; e, Gneisses in the axial zone; f, Pardailhan nappe; g, Mont Peyroux nappe; h, Mont de Faugères nappe; i, Northern cover of the gneisses; j, Autochtonous cover of the gneisses; k, Root zone of the recumbent folds. The Devonian paleomagnetic sampling sites are shown (1 to 8 in the Mont Peyroux nappe, FB in the Bissous nappe is given for completeness but not used in the rest of this study).

strong phase 4 with upright folds trending N60°E and, locally, a weak vertical cleavage (S4). This is a symmetrical final shortening phase which we will term phase B. These phases are supposed to have followed each other more or less continuously in Namurian time (335 to 320 Ma; see Hamet and Allègre [1976], with ages corrected after Steiger and Jaeger [1977]) and to have been accompanied or followed by metamorphism, which led to the emplacement of 320-325 Ma old anatectic granites in the axial zone. Late granites were emplaced about 285-280 Ma [Hamet and Allègre, 1976, age-corrected] and Hercynian tectonics terminated with conjugate NNE-SSW and E-W (brittle) strike-slip faulting lasting until the lower Permian [Arthaud and Matte, 1977; Vernay, 1983]. The shortening direction of these so-called tardihercynian phases [Arthaud, 1970] is actually the same as that of phase B of which they seem to be the last effects. It should be emphasized at this point that the synchronism between the various tectonic phases as observed in different areas of the Montagne Noire [Arthaud, 1970] is open to further testing.

The sites sampled for paleomagnetism were selected on the basis of minimal strain, cleavage and reheating. Some of them (1 and 3) may actually be close to the upper cleavage front of S1 and S3. Sites 1-A, 1-B and 3 are on the northern, quasi-cylindrical, flank of the Roquebrun synform, although their local bedding

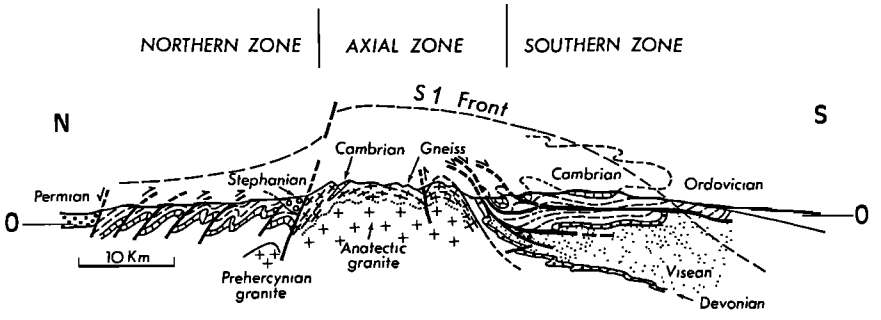


Fig. 1b. Schematic cross-section through the Montagne Noire [after Arthaud, 1970].

attitudes have been made quite different by phase 3 (Figure 1c). Sites 2-A, 2-B, 4, 5, 6 and 7 are close to the eastern synform closure and site 8 is on the southern flank, in an area which has additionally undergone the more recent effects of Pyrenean tectonics. An additional site, FBDS, was sampled for comparison 10 km ENE of the Roquebrun synform in another nappe (Bissous Nappe [Arthaud, 1970]) (Figure 1a); data for this site are provided in the figures and tables, but are not further used in the paper.

Finally, two sites were sampled 15 km to the NE near Lodève, in the lower Permian (Autunian) siltstones, sandy shales and tuffs (Figure 1a). These sites were intended to provide a test for potential younger remagnetizations.

At least 20 and up to 60 cores were sampled in each site with a portable electric drill. Results from independant studies were kindly made available for two sites by W. Lowrie (1-L, 8-L) and for one site by A. Nairn (4-N).

3. MAGNETIC STUDY OF DEVONIAN LIMESTONES

The natural remanent magnetization (NRM) of about 360 samples was measured on three different magnetometers: a Schönstedt DSM-1 spinner, a CTF cryogenic and a Molspin spinner magnetometer. Alternating field (AF) and thermal progressive demagnetizations were done with a Schönstedt single-axis demagnetizer and a laboratory-built, low field furnace respectively, following standard procedures. Details of the measurements are given by Rochette [1981] and Chambon [1984]. NRM intensities ranged from 4.10^{-4} to 6.10^{-1} Am^{-1}, with rather good grouping of NRM directions in sites 1, 2, 3, 4 with evidence of streaking toward the present field direction.

Typical vector plots of AF and thermal

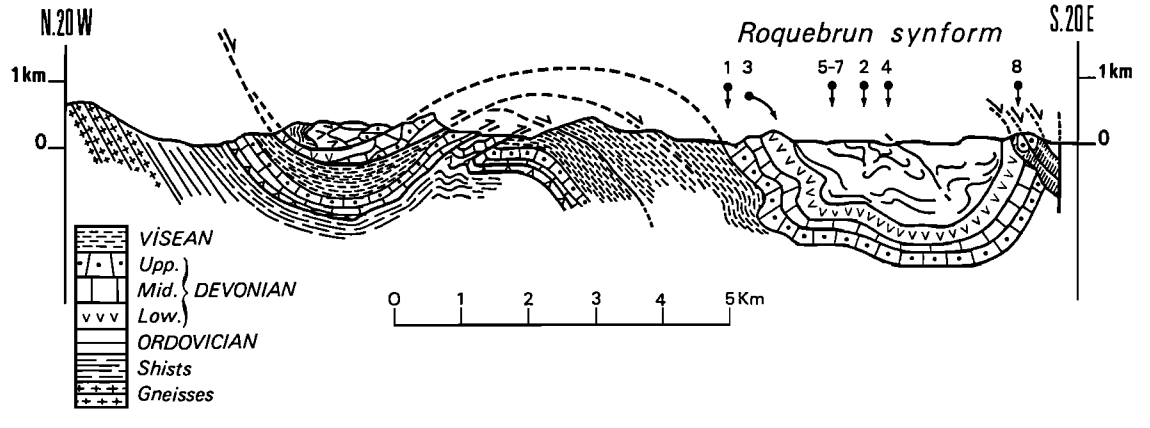


Fig. 1c. Cross section through the Mont Peyroux nappe in the Roquebrun synform area [after Arthaud, 1970]. Location of the section is shown in Figure 1a. Paleomagnetic sites are projected on the section.

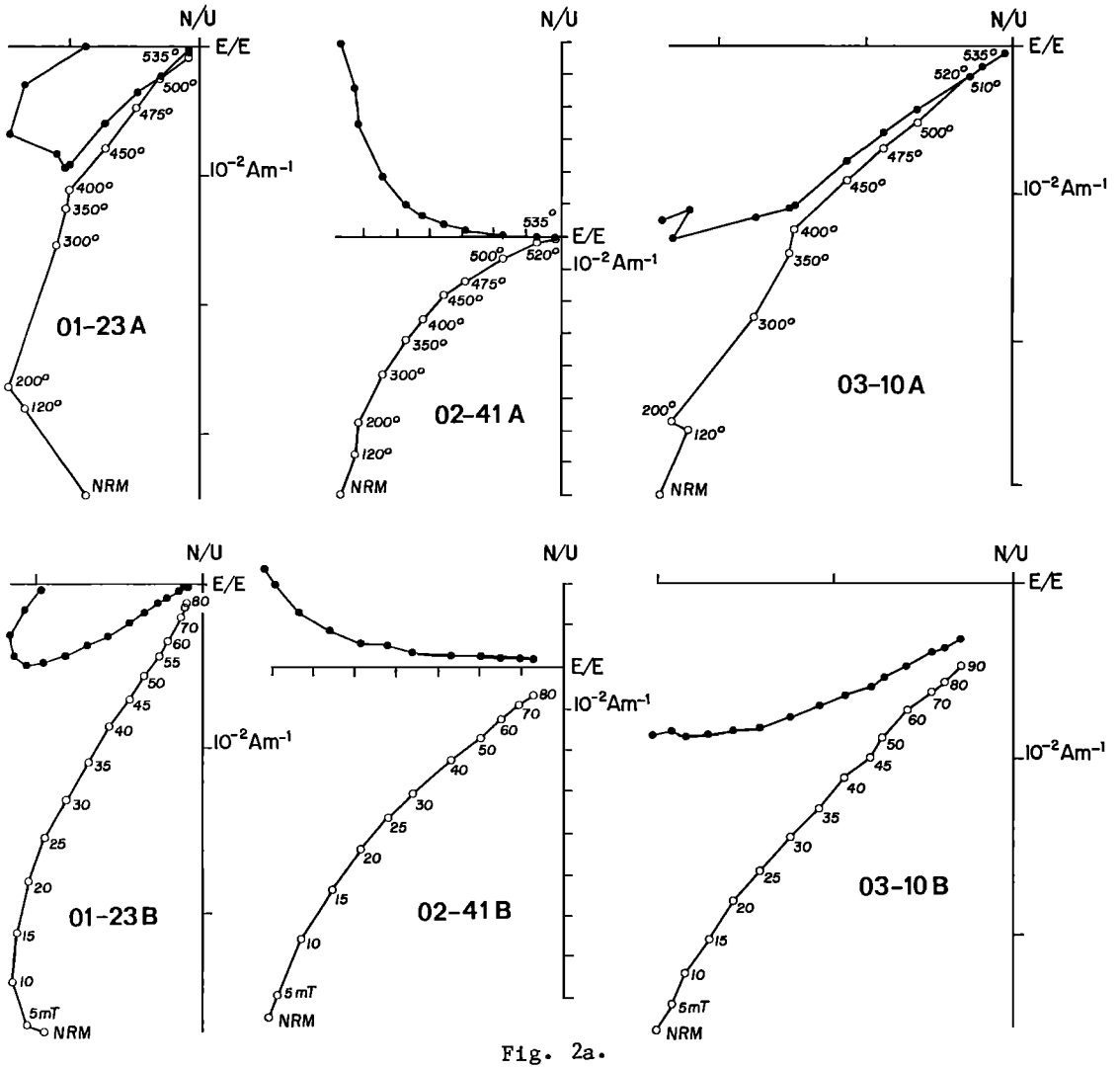


Fig. 2a.

Fig. 2. Demagnetization diagrams for two companion specimens from single cores (top: thermal demagnetization with temperatures in degrees Celsius; bottom: alternating field demagnetization with peak field values in milli-Teslas) taken from six sites of Devonian griotte limestone (1 to 6). All diagrams show horizontal projections (solid dots) and vertical projections in the E-W plane (open dots). Scale of 10^{-2} Am^{-1} is shown.

demagnetizations for paired samples from six sites are shown in Figure 2. Depending on sites and samples, two or three discrete remanences could be isolated. There is generally good agreement between directions revealed by the two techniques. A first dispersed viscous component, close to the present field direction, is removed by about 350° C and 30 mT. A stable characteristic component with unblocking temperatures between 300 and 520° C is clearly apparent in the vector plots. It is slightly less well determined on AF diagrams due to overlapping coercivity spectra of the components, the intensity being reduced to 10 to 20% of the NRM by 80 mT.

A more detailed inspection of the vector diagrams of Figure 2 indeed shows that a dominant component, characterized by low coercivities (less than 80 mT) and blocking temperatures in the range 400-540° C, is sometimes slightly obscured by an overprint with higher coercivities (larger than 80 mT) and relatively low blocking temperatures, between 300° C (sample 04-43A) and 450° (sample 05-85A),

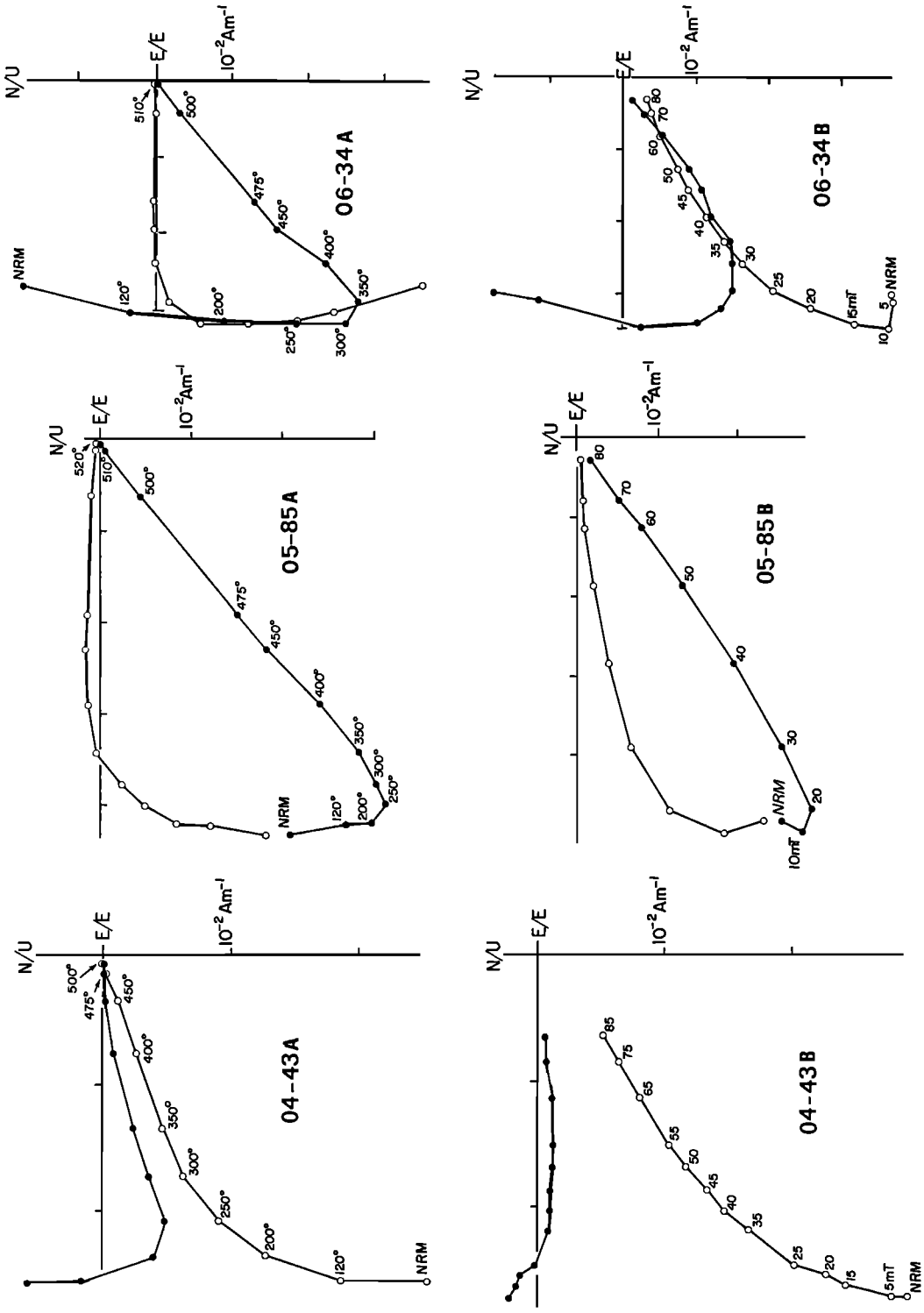

N/U
E/E
$10^{-2}Am^{-1}$
06-34 A
06-34 B
05-85 A
05-85 B
04-43 A
04-43 B
NRM
5mT
10mT
15mT

Fig. 2b.

TABLE 1. Paleomagnetic Directions for Upper Devonian Sites

Site	s	d[g]	N	k	α_{95}	D_g	I_g
1-A[a]	100	48 S	20	332	2.2	231	40
1-B	102	46 S	21	55	4.3	224.7	37.8
1A/1B	101	47 S	2	...	3.4	228	39
1-L[b]	100	45 S	4	453	4.0	217.7	39.0
2-A[c]	207	17 NW	9	43	7.1	269.3	13.7
2-B	345	89 E	9	137	4.4	271.8	36.8
3	281	78 N	13	87	4.5	235.7	36.6
4	7	89 E	24	74	3.5	261.0	25.1
4-N[d]	7	89 E	82	...	...	253	27.5
5	137	53 SW	17	45	5.4	229.4	0.2
6	146	53 SW	8	83	6.1	230.7	-3.9
7	137	64 SW	7	94	6.3	216.0	-4.9
8-A[e]	214	75 NW	8	30	10	188	-26
8-B	208	84 NW	11	19	10.8	278.1	60.7
8-L[b]	210	65 NW	4	12	27.0	215.5	53.3
FBDS[f]	245	20 NW	10	138	3.8	276.9	25.4

Site-mean directions of Characteristic Remanent Magnetization (ChRM) for 15 sites of Devonian griotte limestones from the Mont Peyroux nappe (Montagne Noire). Mean specimen directions are determined by least squares [Kirschvink, 1980] unless otherwise specified. Columns give site identification, strike (s) and dip (d) of bedding, number of specimens (N), Fisher's precision parameter (k) and 95% confidence angle (α_{95}), mean declination (D_g) and inclination (I_g) in present-day geographic coordinates. Angles in degrees. Site FBDS is included for completeness but being the only one in the Bissous nappe, a separate tectonic unit, it is not used further in this study.

a Directions at 420° C.
b W. Lowrie, (personal communication, 1983).
c Directions at 320° C.
d A. Nairn, (personal communication, 1984).
e Directions at 530 to 580° C.
f Directions at 500° C; site at Pic de Bissous.
g All sites overturned except 2-B, 3, 4, 4-N.

more often around 400° C (samples 01-23A, 03-10A, 06-34B). The dominant component is carried by some form of (SD) magnetite, as confirmed by further analyses by Rochette [1983, pp. 53-56]. For whole rock samples from site 1-A, Rochette finds low field ferromagnetic specific susceptibilities on the order of 5 10^{-8} $m^3 kg^{-1}$, high field susceptibilities of 3 10^{-8} $m^3 kg^{-1}$ and saturation induced specific magnetizations of 3.10^{-3} $Am^2 kg^{-1}$. For some samples, 95% of saturation IRM is reached by 400 mT at 300K and the characteristic Verwey transition at 118K is observed on heating from 10K. The secondary (minor) component is clearly carried by hematite. Corresponding samples show a pink pigmentation and higher coercivities. For a more coloured sample from site 1-A, 95% of saturation IRM is reached only after 3000 mT at 300K.

Following the pilot study, all samples were demagnetized at 6 or 7 temperature steps from 200 to 520° C. Only site 1 provided better determinations with AF demagnetization and 6 steps between 20 and 80 mT were selected. Characteristic remanent magnetizations (ChRM) were determined by least squares [Kirschvink, 1980] in a selected thermal or AF unblocking range (400-500° C; 30-80 mT depending on the site). Samples from site 8 had abnormal behaviour, with weaker NRM, higher unblocking temperatures and evidence for secondary alteration (strong red pigmentation). This site is in a more disturbed part of the synform, close to the southern

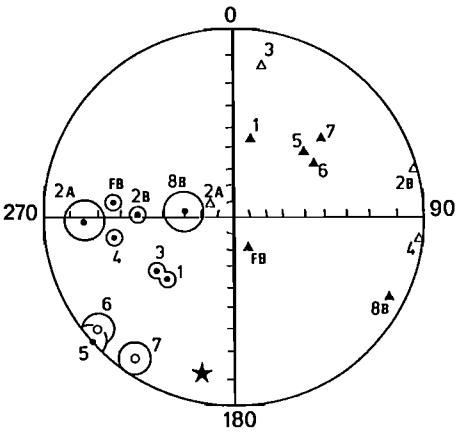


Fig. 3. Site-mean directions of characteristic remanent magnetization (ChRM) with their α_{95} oval of confidence and poles to bedding (directed towards the younger strata) at each site are shown respectively as circles and triangles (see also Table 1). The paleomagnetic direction expected if the sites were part of stable Europe in the upper Carboniferous is shown as a star (see Table 2 and text). Closed (open) symbols are in the lower (upper) hemisphere. In situ coordinates.

edge of the Hercynian nappes, where Pyrenean tectonics might have affected the Paleozoic strata.

Site mean directions for between 7 and 24 demagnetized samples (depending on the site) are given in Table 1. Except for site 8, they all give k values larger than 40 and α_{95} smaller than 7°. Site 1-L was sampled very close to our subsites 1-A and 1-B by W. Lowrie (personal communication, 1983), and the results are in excellent agreement. Site 4-N was sampled for magnetostratigraphy by A. Nairn (personal communication, 1984) at our site 4 and again there is excellent agreement. Subsites 8-L (W. Lowrie, personal communication, 1983), 8-A and 8-B give divergent results, another indication of the tectonic and chemical disturbances suffered in this area. Site 8-B is in a quarry which seems less affected by alteration (colour) and tectonics, and will be used (again with caution) in the discussion of the results.

4. MAGNETIC RESULTS FOR DEVONIAN LIMESTONES

Mean values and uncertainties on ChRM directions are plotted in Figure 3 in present (in situ) geographic co-ordinates. Although mean directions from each site are well determined, they display a large between-site scatter. Yet, none of them is close either to the present field direction or to any direction expected for this area since Mesozoic time. Thus, site-mean ChRM directions should have retained a magnetic memory that must predate the latest major tectonic phase (i.e. Hercynian phase B). If a classical tilt correction is applied, the ChRM directions remain highly scattered (Figure 4). The fold test is negative according to McElhinny's [1964] statistics and it is likely that the original Devonian magnetization has not been preserved. Returning to the in-situ directions (Figure 3), we find that they are not randomly distributed but vary in a rather regular manner as one follows the Devonian beds around the synform. Sites 1 and 3 on the northern limb provide essentially identical directions despite rather different bedding attitudes, sites 5 to 7 close to the hinge are well grouped, and remaining sites 2-A, 2-B and 4 from the south-east side of the hinge zone are also well grouped. Therefore, there is a logic in the distribution of directions which seems to be related to the overall present shape of the synform. The simplest (and testable) hypothesis, in the view of these magnetic results (in particular the small within-site and large between-site scatter) and of the previous summary of tectonic phases in the Montagne Noire, is that of a complete remagnetization occurring between tectonic phases. The phases postdating the remagnetization would then be

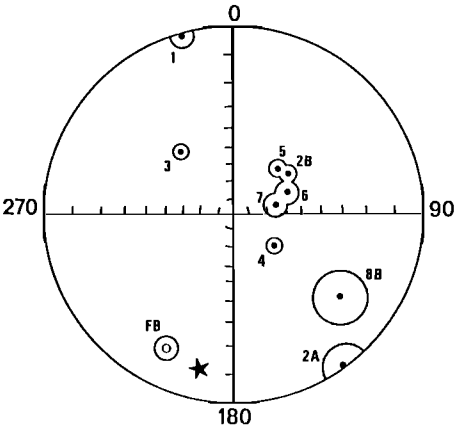


Fig. 4. Site-mean directions of ChRM after restoring the bedding planes to the horizontal along their line of strike (classical "tilt-correction"). Symbols as in Figure 3.

responsible for the scatter of observed ChRM.

Magnetic anisotropy measurements have been carried out on a sample from site FB. After saturation in a field of 4000 mT amplitude, the anisotropy of isothermal remanence reached 7.5%, being maximum in the bedding-cleavage plane. In case of a remagnetization following the acquisition of anisotropy, this would lead to a maximum angular deviation of 5° of the ChRM [Rochette, 1983]. This means that anisotropy can be safely neglected in our further discussion of the remagnetization.

5. AGE OF THE REMAGNETIZATION

Several lines of evidence allow us to place strong constraints on the age of the widespread remagnetization.

1. We can first show that the remagnetization must postdate phase 3. Sites 1 and 3, on the northern E-W trending limb of the synform, are on the two limbs of a small phase 3 fold which locally complicates the structure of the recumbent synform [Arthaud, 1970] (Figure 1b) and have very different bedding attitudes (Table 1). Yet, sites 1 and 3 have very similar ChRM. Even though we have only two sites, we can attempt a fold test and find a k ratio of 48, implying a negative fold test at the 95% confidence level [McElhinny, 1964]. Similarly, sites 2-A, 2-B and 4 are part of phase 3 minor folds close to the hinge and the k ratio for the corresponding ChRM is 15, implying a negative fold test at the 99% confidence level. Actually, as said earlier, tectonic phase 3 is but the final event of the main phase A and remagnetization must postdate this phase. According to Arthaud [1970] and Hamet and Allègre [1976], anatectic granites are emplaced in the axial zone to the north between 320 and 330 Ma. It is thus reasonable to assume that remagnetization is linked to this major thermal event.

2. Remagnetization must predate phase B since this is the only tectonic phase of sufficient amplitude to disperse the ChRM in the way which is observed (Figure 3). Phase 4 is not accurately dated, and Arthaud et al. [1970] and Hamet and Allègre [1976] assume that it follows phase A without interruption and is coeval or even older than the anatectic granites (~325 Ma). This is not really compatible with the timing of remagnetization and phase B must be somewhat younger than thought earlier. However, stratigraphic unconformities (Stephanian over Visean) place an upper limit of about 300 Ma on the age of phase B (Figure 9).

3. We now know that the remagnetization must be a Carboniferous event. The paleomagnetic data do not show a single reversal and directions tend to be in the SW quadrant (Figure 3), which suggests that the remagnetization must have occurred during the Permo-Carboniferous Kiaman reversed superchron [Harland et al., 1982]. This fixes a lower boundary of about 320 Ma for the remagnetization.

These various lines of evidence show that the remagnetization event must have taken place between 320 and 300 Ma, and have been followed by phase B also prior to 300 Ma. We can thus estimate that remagnetization occurred around 310-320 Ma and that phase B took place around 300-310 Ma. According to Hamet and Allègre, metamorphic temperatures are still about 500° C at that time in the axial zone close to the newly emplaced anatectic granites. It is unlikely that the remagnetization can be associated with the late (280 Ma) granites (or slightly younger mineral ages around 270 Ma; Hamet and Allègre [1976]) because it would make phase B far too young, in contradiction with observed Carboniferous unconformities, and also because the late Hercynian brittle tectonics alone cannot account for the observed scatter of NRM directions with respect to the expected upper Carboniferous direction of the field. Moreover, the ~280 Ma lower Permian directions are preserved, as shown in the next section.

6. MAGNETIC STUDY OF LOWER PERMIAN SITES

As a check for the lack of any sizeable post-Carboniferous thermal event, we have sampled two sites in the lower Permian (Autunian) of the Lodève basin, lying unconformably over the Stephanian of the Montagne Noire, 15 km to the NE of the Roquebrun sites. Tuffs and sandy shales from an Uranium mine (site FUPA) and siliceous shales from a second nearby site (FMPA) were demagnetized. Both AF and thermal procedures reveal that magnetite is a likely carrier of a well determined ChRM. Unblocking temperatures are in the range of 400° to 550° C and thermal treatment provides better estimates of the ChRM. Low unblocking temperature overprints remain in the AF demagnetization diagrams up to 80 mT. All types of lithologies provide the same ChRM, all direc-

TABLE 2. Paleomagnetic Permian (Autunian) Directions Near Lodève

(a)	Site	N	D_g	I_g	K_g	α_{95}	s	d	D_s	I_s	K_s	α_{95}	K_s/K_g
FUPA	cinerites	5	191.3	3.1	24	15.8			191.1	-12.4	54	10.5	
	sediments	3	189.0	19.7	449	5.8	} 8.5	27 S	187.3	-5.1	648	4.8	
	mean	8	190.5	9.4	28	10.5			189.7	-9.7	73	6.5	2.6
FMPA	siliceous shales	3	190.8	13.3	407	6.1			191.0	-4.7	424	6.0	
	sil.sh.+black sh.	7	187.6	23.5	15	15.8	} 108	18 S	187.1	-3.5	59	7.9	3.8
FUPA + FMPA		2/15	189.0	16.5					188.4	-6.6			

(b) References	D_g	I_g	D_s	I_s
Kruseman [1962]	190	21	190	4

(c) References	D_s	I_s
Irving [1977] - 270 Ma	191	6
Zijderveld et al. [1970]	196	-6
Westphal et al. [1986]	191	4

Mean directions of ChRM for two sites of Permian (Autunian) sediments near Lodève. Symbols as in Table 1; g stands for geographic and s for stratigraphic (classical tilt-correction) coordinates. Part b shows the Permian results of Kruseman [1962] obtained with AF demagnetization only. Part c shows directions from various upper Carboniferous poles for stable Europe, predicted at a common site at latitude 43.7°N, longitude 3.3°E.

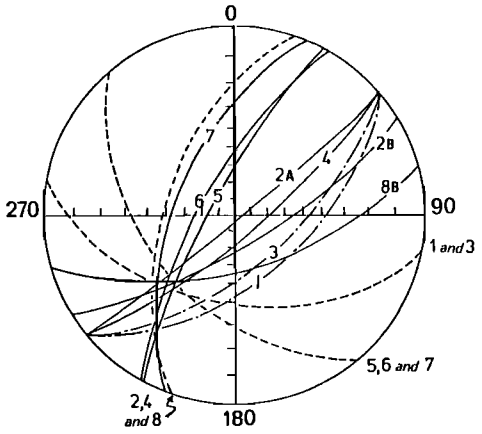


Fig. 5. The loci of rotation axes that bring the ChRM directions observed at each site on the expected (stable Europe) upper Carboniferous direction (see Figure 3) are the great circles bisecting these two directions at each site. These great circles are shown as solid curves for sites close to the hinge of the Roquebrun synform (2A, 2B, 4, 5, 6, 7, 8B) and are seen to intersect at a point close to azimuth = 230°, dip= 30° (point α in Figure 6). The great circles shown as long/short dashed curves are for sites from the northern flank of the synform (1, 3) and are seen to intersect close to strike = 235°, dip = 0° (point β in Figure 6). An average bedding plane is shown for several subgroups of sites as a short dashed curve with the site numbers given on the edges: these averages smooth out small-scale, local (phase 3) folds.

tions are reversed, and despite the moderate number of samples (15), small number of sites (2) and moderate amount of folding, there are indications of positive fold tests both within each site and between the two sites (Table 2). This yields a good estimate of the lower Permian field direction and indeed it is very close to the corresponding directions computed from poles for stable Europe [Irving, 1977; Zijderveld et al., 1970; Westphal et al., 1986]. The Zijderveld et al. [1970] estimate is closest to our observed direction. Note a good agreement with the direction found by Kruseman [1962] for other lower Permian sites in the Lodève basin. The direction found by Kruseman relies entirely on AF demagnetized samples and is virtually identical to the one we find also with AF treatment for our samples. However, as mentioned above, these are not completely devoid of an overprint with low unblocking temperature and moderate coercivity. Thus, our estimate of the Autunian direction (188°, -7°) is likely to be a better estimate for the lower Permian of the Lodève basin. Parenthetically, it implies that the Montagne Noire-Lodève area had already crossed the equator by Autunian time, as suggested by the Zijderveld et al [1970] pole, but not by the Irving [1977] and Westphal et al. [1986] poles. The Permian magnetization of the Lodève basin has not been significantly erased by the ~170 Ma old 100°-200° C thermal event which Lancelot et al. [1984] find in the uranium bearing Autunian pelitic layer and which they associate with a Jurassic distensive phase, linked to the opening of the Liguria and Piemont basins (see also Maury and Mervoyer [1973]). In any case, this side study demonstrates that an Autunian magnetic memory has been preserved close to the Roquebrun area (admittedly in quite different lithologies) and that the Roquebrun magnetization is likely to predate it.

7. UNFOLDING MODEL

Since we have a rather precise estimate of the age of remagnetization, we can com-

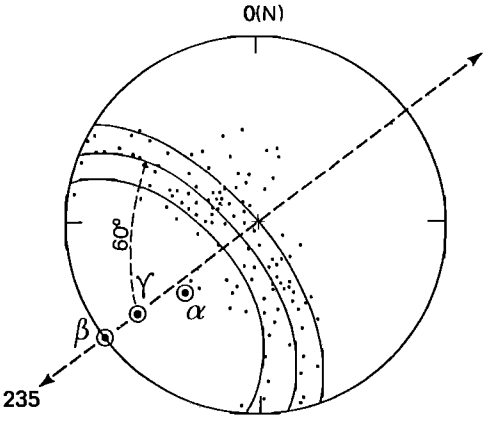


Fig. 6. Small dots represent poles to the axial planes of 125 minor folds generated in the Roquebrun synform during tangential phase A [phase 1 of Arthaud] and later dispersed along a small circle by phase B (phase 4 of Arthaud) which generated the synform (after Arthaud, 1970). The axis of this small circle (γ) has approximately the same azimuth as the axes (α) and (β) determined magnetically from Figure 5, and an intermediate (average) dip.

TABLE 3. Unfolding of Devonian Directions

Site	Pole to Bedding (in Situ)		Small Circle Rotation			Pole to Bedding (Rotated)		Angle (1 vs 2) A	Magnetic Direction (Rotated)	
	az	d	az	d	ω	az	d		D	I
1	11	43	230	-6	60	79	50	45	190	15
2A	117	73	230	35	-150	-87	-13	61	186	20
2B	75	-1	230	35	-150	-2	-38	79	207	9
3	11	-12	230	-6	60	12	31	43	194	19
4	97	-1	230	35	-150	-9	-17	75	198	27
5	47	37	230	35	-50	-2	17	47	203	10
6	56	37	230	35	-50	3	22	47	202	6
7	47	26	230	35	-50	5	9	43	189	14
8B	118	6	230	35	-150	-58	-47	41	194	12

Restoring the ChRM directions listed in Table 1 (D_g, I_g) using the unfolding model determined from Figure 5 (and 6). The azimuths az and dips d of poles to bedding (columns 2 and 3) correspond to the strikes s and dips d in Table 1. The elements of each rotation are given in columns 4 to 6 (azimuth az and dip d of the rotation axis, and total angle of rotation ω). The rotated poles to bedding and ChRM directions are listed respectively in columns (7, 8) and (10, 11) and shown in Figure 7. The great circle angle A between the present-day (columns 2, 3) and restored (columns 7, 8) poles to bedding is given in column 9.

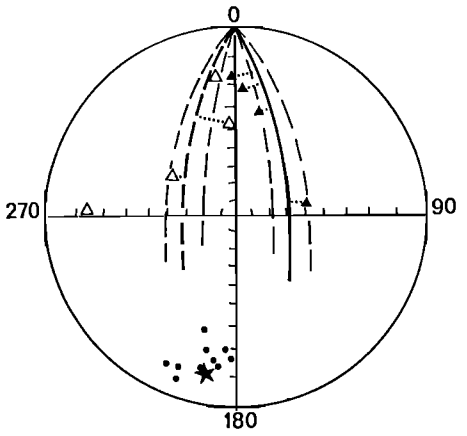


Fig. 7. Magnetic and structural directions of Figure 3 are "unfolded" along the axes α and β (depending on the site) determined in Figures 5 and 6. The "unfolding" parameters are given in Table 3. Symbols are as in Figure 3. The magnetic directions cluster near the expected upper Carboniferous direction and the poles to bedding are not far from a great circle girdle, indicated as a bold solid (respectively, dashed) line in the lower (respectively, upper) hemisphere. Only one site (2-A) fails this. The normal to the girdle dips 30° due west.

pute from stable European poles at the time [Irving, 1977] the paleomagnetic direction which should have been observed at the Roquebrun sites. This is shown as a star in Figures 3 and 4. For a given site, the observed direction can be made to coincide with the predicted one by a single rotation about an infinite number of possible axes whose locus is the great circle that bisects the two magnetic directions. Such a great circle can be determined for each site as is shown on Figure 5. Great circles from all sites (except 1 and 3) are found to converge to a single well determined point. This means that a single rotation axis can be found to bring all observed directions from sites near the hinge of the synform to the expected middle Carboniferous direction. The azimuth of this rotation (230°) is very close to the fold axis direction of phase B and bissects the conjugate sets of tardihercynian faults. Figure 6 [from Arthaud, 1970] shows that the axial planes of small phase 1 (or A) folds within the Roquebrun synform form a small circle girdle of 60° aperture around a plunging axis which has exactly the same azimuth as our rotation axis: this is the average phase B (plunging) axis. On the other hand, the great circles corresponding to sites 3 and 1 on the northern limb of the synform intersect close to the horizontal plane, again very close to the 230° azimuth (Figures 5 and 6). A rotation around this direction will bring ChRM of those sites on the expected Carboniferous direction.

Once the rotation axes have been determined (Figure 5 and Table 3, columns 4 and 5), the total angle of rotation ω required to bring the observed and expected magnetic directions for each site in coincidence can be evaluated (Figure 5 and Table 3, column 6). Assuming that no significant internal deformation has occurred, the ChRM directions should behave as material lines and the rotations we just determined must be those that deformed the post-phase A structures to generate their present post-phase B geometry. The rotations can then be undone to restore the pre-phase B geometry. This is done in Figure 7 (also Table 3, columns 7 and 8) where the in situ poles to bedding of Figure 3 (also Table 3, columns 2 and 3) have been rotated back, at the same time restoring the paleomagnetic directions to their expected Carboniferous direction (also Table 3, columns 10, 11). Except for one site (2-A) which is closest to the hinge zone, the eight other poles to bedding cluster close to a great circle striking due north (in present day coordinates) and plunging about 60° to the east. This describes the structural features at the end of tectonic phase A, with approximately cylindrical E-W folds plunging about 30° to the west.

The unfolding model will be of some value if it is both simple enough and can be interpreted in mechanical terms. For instance, one can always find for each site a rotation that will bring the ChRM directions to the expected direction (such as the great circle rotation), but the resulting set of rotations will not bear relation with the strain pattern along the synform. The rotations we have determined are far simpler, all of them having the same azimuth which is virtually identical with the azimuth of phase B folding (Figure 6). However, we see from Figure 5 that a single rotation axis is not sufficient and that two separate axes must be used, one for sites (1,3) from the northern limb of the synform, the other for sites (2-A, 2-B, 4, 5, 6 and 7) close to the eastern hinge zone. The relationship between these two axes and the fold geometry can be

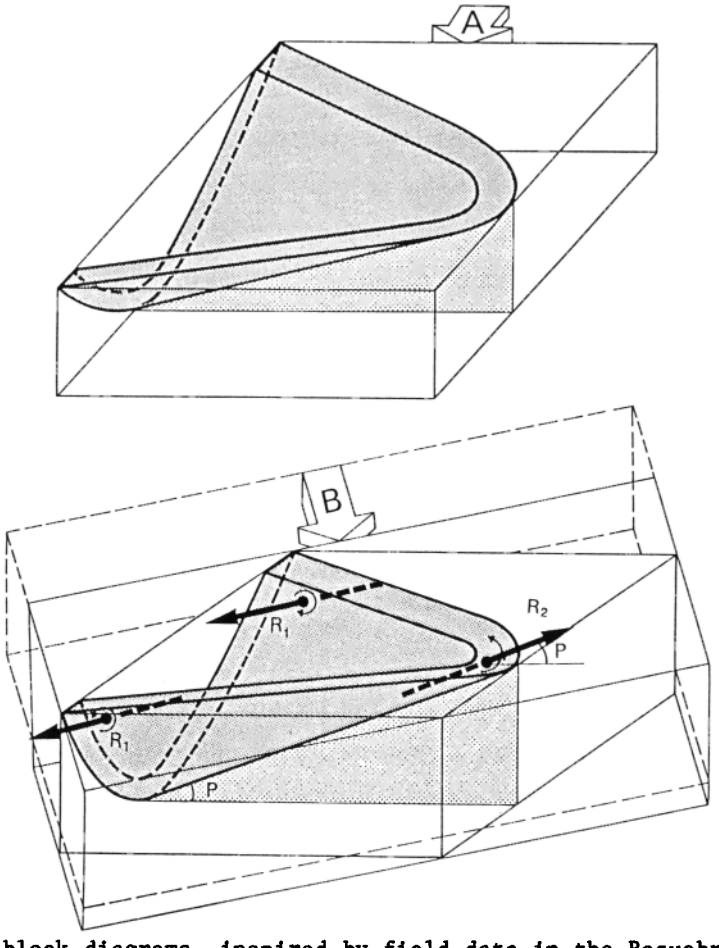


Fig. 8. Two block diagrams, inspired by field data in the Roquebrun synform and buckling experiments of Ghosh and Ramberg [1968], show how structure inherited from phase A (top) influences the development of phase B folds. The material far from the hinge of the synform is refolded around horizontal (R_1) axes perpendicular to the B shortening direction, as expected, whereas material close to the hinge rotates around it and is folded around dipping (R_2) axes, guided by the hinge line. Phase B tightens the overall structure and increases the plunge.

understood in the light of experiments by Ghosh and Ramberg [1968]: these authors describe buckling experiments on intersecting fold patterns and show that, when the angle of a second phase of folding is about 30° or less to the previous phase, the folds produced in the first phase will control the development of phase B folds. Indeed, the magnetic data presented here (and also the structural data from Arthaud [1970] to some extent) show that the fold axis direction (and presumably the principal horizontal stress components) have rotated about 30° counterclockwise between phase A (Figure 7) and phase B (Figures 5 and 6). Figure 8 attempts to illustrate qualitatively the results of the Ghosh and Ramberg [1968] experiments. The upper block diagram shows part of a plunging synform due to phase A. The lower block diagram shows the ensuing strain after phase B, which is at an angle of about 30° with respect to phase A. The plunge of the synform is increased and the synform is further tightened. Far from the hinge zone, the flanks of the synform are rotated in opposite senses about horizontal axes R_1, perpendicular to the phase B direction of compression. Small folds can be produced locally in these flanks with

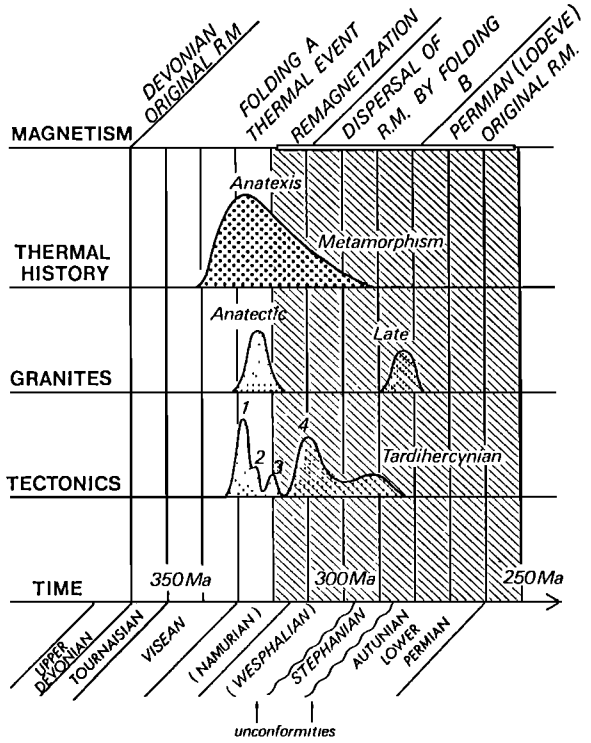


Fig. 9. Summary of events in the Mont Peyroux nappe and Montagne Noire. The geological time scale and local unconformities at the bottom (the Namurian and Westphalian are actually not observed and the Stephanian unconformably overlies the Visean; Engel et al. [1982]) and intensity of tectonic phases (with their names) above them are after Arthaud [1970]. The thermal history and ages in the axial zone to the north are after Hamet and Allègre [1976], with ages corrected after Steiger and Jaeger [1977]. The sequence of magnetization events provided by this paper is shown on top. The main modification to earlier studies provided by the magnetic analysis is that tectonic phase B (phase 4 of Arthaud) occurs around 310 Ma, postdating the emplacement of anatectic granites, postdating the beginning of the Kiaman reversed superchron (ruled zone from 250 to 320 Ma) and predating the unconformity of Stephanian over Visean. The age of remagnetization is 310-320 Ma, prior to phase B.

R_1 axes, since the curvature of the beds away from the hinge is small. In conclusion, phase B folds will develop relatively unhindered far away from the hinges of phase A folds. We suggest that this situation occurs for sites 1 and 3. On the other hand, the curved beds close to the hinge zone will interfere with phase B deformation, forcing material to rotate around the hinge, more or less about a rotation axis R_2 locally parallel to the hinge. We suggest that this situation holds for sites 2, 4, 5, 6 and 7.

Once the axis of rotation (R_1 or R_2) has been determined, the actual amount of rotation will depend on the position of the site with respect to the hinge and on the material properties of the competent layers and of the surrounding less competent strata: we know of no quantitative treatment of this problem. From our analysis of ChRM directions, we find a ~50° rotation for sites 5, 6 and 7 on the northern side of the hinge and ~-150° for sites 2-A, 2-B, 4 and 8-B on the southern side. A discussion of these high values would require further detailed microtectonic fieldwork.

8. EXTENT AND ORIGIN OF THE REMAGNETIZATION

Our data reveal that the Mont Peyroux nappe, and possibly all of the Montagne Noire, suffered widespread total remagnetization at about 310-320 Ma, that is in middle Carboniferous time (Figure 9). The sediments of the various nappes south of the axial zone of Montagne Noire are generally regarded as nonmetamorphic, at most in the chlorite epizone [Arthaud, 1970]. This is stated for the Devonian griottes of the Mont Peyroux nappe by Feist [1983]. A study of fluid inclusions in quartz veins along the basal contact of the Pardailhan nappe over the Mont Peyroux nappe (5 to 10 km west of our sites, see Figure 1a) leads to a local maximum temperature of 275° C [Guiraud et al., 1981]. This temperature is related to shearing tectonics at the base of the nappe and is likely to be higher than that suffered by the sites we sampled. It is related to phase A [Guiraud et al., 1981]. Thus, although the remagnetization is clearly associated to a tectonothermal event, it does not seem to be a thermal but rather a chemical remagnetization, primarily carried by SD magnetite. The hematite overprint is minor and post phase-B, and could also partly be related to much younger Pyrenean tectonics. The Hercynian remagnetization could be due to widespread circulation of chemically active groundwater following the mechanical and thermal peaks of phase A in the axial zone.

As recalled by Irving and Strong [1985,

p.1959], widespread Kiaman overprinting is characteristic of the Paleozoic record in Europe [Creer, 1968]. In a recent review of Devonian and Carboniferous paleomagnetic data from Hercynian Europe, 12 out of 18 results are middle to upper Carboniferous remagnetizations in the Armorican Massif, and Iberian Meseta [Perroud, 1986]. For instance, Edel and Coulon [1984] isolate a Westphalian and a latest Carboniferous-earliest Permian overprint in lowermost Carboniferous sediments from the Armorican Massif. Salmon et al. [1985] describe late Carboniferous to Permian overprinting of middle to late Devonian limestones from southern Morocco, which can be directly correlated with the Montagne Noire series. Blocking temperatures are also in the 450-580° C range. Robinson [1985] finds a late Paleozoic remagnetization in upper Middle Devonian sediments from northern Scotland. He interprets this as a pervasive chemical remagnetization carried by hematite and due either to oxidation of detrital titanomagnetite, oxidation of biotite or dedolomitization of ferroan dolomite, all linked to deep weathering in the upper Paleozoic. These are not limited to Europe: McCabe et al. [1984] demonstrate the complete remagnetization of middle Ordovician limestones from Quebec and New York State in the late Carboniferous-early Permian. The ChRM is carried by magnetite in the 350-550° C range and is thought to be a chemical (diagenetic) remagnetization caused by "deep basinal fluids migrating towards the craton during Alleghenian mountain building." Even more recently, Irving and Strong [1985] find a chemical overprint in the St Lawrence granite in Newfoundland acquired during the Kiaman reversed superchron. Moreover, Irving and Strong convincingly argue that most middle Devonian to lower Carboniferous paleomagnetic results from the North American craton appear to be reversed Kiaman overprints. These overprints are apparently not found in Precambrian rocks, suggesting again to Irving and Strong a comparatively superficial chemical event related to late Paleozoic groundwater circulation.

The rather precisely dated middle Carboniferous total remagnetization in the Montagne Noire (Figure 9) thus appears to be but an instance of a continent wide remagnetization event that wiped out the magnetic memory of most earlier rocks, particularly Devonian ones. Most rocks of that age coming from Laurasia, Baltica, Armorica and northern Gondwana bear witness to this major event. The carrier of ChRM often appears to be magnetite with blocking temperatures between 400 and 500° C and it can be shown in many instances that the remagnetization is not a thermal event, despite close time association with Hercynian tectonics. A chemical event linked with groundwater circulation is often invoked, although the details of the process are far from understood. A dolomitization-dedolomitization process, as pointed out by M. A. Robinson (personal communication, 1984), might be important in this respect. Although all of our samples came from neritic to pelagic limestones free from dolomitization, it should be pointed out that dolomitization does affect the lowermost part of the Devonian series [Feist, 1983]. Further studies in magnetic mineralogy of the carrier of the overprint, detailed age determinations and sedimentological studies will be needed to advance understanding of this most remarkable event.

9. CONCLUSION

It has been shown that Devonian griotte limestones from the Montagne Noire were totally remagnetized around 310-320 Ma between the two major Hercynian tectonic phases that shaped the Massif (Figure 9). The remagnetization is carried by magnetite in the 400°-540° C range of blocking temperatures and appears to be a chemical remagnetization. It may be an example of a global Carboniferous remagnetization event that can be traced throughout much of Laurasia, Armorica and possibly Baltica and Gondwana. The accurate determination of the age of remagnetization allows a significant number of tectonic inferences to be drawn from the magnetic data. The shortening directions and plunges of the two major tectonic phases can be independantly determined both in present-day and original geographic coordinates. The shortening direction for the Namurian phase A was about N10° W, folding was rather cylindrical and plunged 30° to the West. Phase B occurred later than originally thought. It was mostly a Westphalian event, with a shortening direction of N40° W at the time. Thus, the shortening direction rotated clockwise by about 30° in the course of ~10 Ma, corresponding to different stages in the process of collision of Gondwana against Laurasia-Armorica (for instance, Perroud [1986]). Rotations of

this order of magnitude have been found in Tibet [Achache et al., 1984] and Thailand [Achache and Courtillot, 1985] to be associated with the rotation of India as it collided against Eurasia (see also Patriat and Achache [1984]). Rotations have also been inferred to be associated with Hercynian tectonics, at various scales [Edel and Coulon, 1985; Perroud, 1986; Van der Voo and Channell, 1980].

Finally, the simple geometric construction which allows tectonic elements of phase B to be deduced from a comparison of observed and predicted magnetic directions at the time of overprinting demonstrates that the phase A folds control the development of phase B folds, in agreement with some experimental results on interacting fold patterns. This is the reason why the unfolding model cannot be reduced to a single rotation but requires different rotations depending on whether sites are located far from or close to hinge zones of phase A folds. The unfolding of phase B finally allows one to restore the geometry of deformation resulting from phase A alone.

This study demonstrates the potential of remagnetization studies for the solution of complex tectonic problems. A rather limited amount of paleomagnetic sampling, reinforced by pre-existing microtectonic and geochronologic data forms the basis of this, and any future application of magnetotectonics.

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